

Medical Procedures & Diagnostics

Clinical Investigations, Interpretation & Procedural Guidance

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1. Laboratory Investigations

1.1 Complete Blood Count (CBC/FBC)

The complete blood count is one of the most commonly ordered laboratory tests, providing quantitative information about erythrocytes, leukocytes, and platelets. Interpretation must always be in clinical context.

Parameter	Normal Range (Adult)	Low Significance	High Significance
Haemoglobin	M: 130–175 g/L F: 115–160 g/L	Anaemia (classify by MCV)	Polycythaemia, dehydration
WBC	4.0–11.0 $\times 10^9/L$	Neutropenia (infection risk)	Infection, inflammation, leukaemia
Neutrophils	1.8–7.5 $\times 10^9/L$	Neutropenia ($< 0.5 =$ severe)	Bacterial infection, stress, steroids
Lymphocytes	1.0–4.0 $\times 10^9/L$	Viral infection, immunodeficiency	Viral syndromes, CLL
Platelets	150–400 $\times 10^9/L$	Thrombocytopenia: bleeding risk	Thrombocytosis: reactive or essential
MCV	80–100 fL	Microcytic: Fe, thalassaemia	Macrocytic: B12/folate deficiency

1.2 Comprehensive Metabolic Panel

The metabolic panel assesses renal and hepatic function, electrolyte balance, and glucose homeostasis. Results must be interpreted together — no single value tells the whole story.

Test	Reference Range	Clinical Significance of Abnormality
Glucose (fasting)	3.9–5.5 mmol/L	High: DM, stress; Low: insulinoma, drugs, sepsis
Urea (BUN)	2.5–7.1 mmol/L	High: AKI, CKD, GI bleed, dehydration, high protein diet
Creatinine	M: 62–115 $\mu\text{mol/L}$ F: 53–97 $\mu\text{mol/L}$	High: AKI/CKD — track trends for GFR estimation

Test	Reference Range	Clinical Significance of Abnormality
Sodium	135–145 mmol/L	Hypo: SIADH, HF, cirrhosis; Hyper: dehydration, DI
Potassium	3.5–5.0 mmol/L	Hypo: diuretics, vomiting; Hyper: AKI, ACEi, haemolysis
ALT	7–56 U/L	High: hepatocellular damage (hepatitis, NAFLD, drugs)
ALP	44–147 U/L	High: cholestasis, bone disease, pregnancy
Bilirubin (total)	< 21 µmol/L	High: haemolysis (indirect) or cholestasis (direct)

2. Cardiac Investigations

2.1 Electrocardiogram (ECG) Interpretation

The 12-lead ECG provides electrical information from multiple anatomical perspectives and is indispensable for diagnosing arrhythmias, ischaemia, and structural abnormalities.

Systematic ECG Reading Approach

- Rate: Count R-R intervals (300/large squares or 1500/small squares)
- Rhythm: Regular vs irregular; identify P waves and relationship to QRS
- Axis: Check leads I and aVF; normal -30° to $+90^{\circ}$
- P waves: Present, morphology, PR interval (120–200 ms)
- QRS: Duration (< 120 ms), morphology, bundle branch blocks
- ST segment: Elevation (STEMI, pericarditis) or depression (ischaemia, digoxin)
- T waves: Inversion (ischaemia, strain, PE), peaked (hyperkalaemia)
- QTc: Correct for heart rate; prolonged (> 450 ms men, > 460 ms women) — arrhythmia risk

■ **STEMI criteria: ST elevation ≥ 1 mm in ≥ 2 contiguous limb leads, or ≥ 2 mm in ≥ 2 contiguous precordial leads. Activate cardiac catheterisation laboratory immediately — call emergency services.**

2.2 Cardiac Biomarkers

Marker	Rise	Peak	Normalise	Significance
Troponin I/T (hs)	1–3 h	12–24 h	7–14 days	Gold standard for AMI; also in PE, myocarditis
CK-MB	3–6 h	18–24 h	48–72 h	Re-infarction detection (rises after normalisation)

Marker	Rise	Peak	Normalise	Significance
Myoglobin	1–2 h	4–6 h	24 h	Early marker; non-specific (also muscle injury)
BNP / NT-proBNP	Hours	Variable	Variable	Heart failure; correlates with severity and prognosis

3. Imaging Modalities

3.1 Chest X-Ray (CXR)

The postero-anterior (PA) chest radiograph is a first-line imaging tool for thoracic pathology. A systematic review using the mnemonic ABCDE ensures no finding is missed.

- A – Airways: Tracheal position, carina angle (normally < 70°), bronchi
- B – Bones: Rib fractures, lytic lesions, clavicles, thoracic vertebrae
- C – Cardiac: Cardiothoracic ratio (normal < 0.5 on PA), borders, aortic knuckle
- D – Diaphragm: Hemidiaphragm levels, costophrenic angles (blunting = effusion ≥ 200 mL), free air
- E – Everything else: Lung fields (consolidation, nodules, pneumothorax), hila, soft tissues, devices

3.2 CT Scan

Computed tomography provides cross-sectional imaging with superior soft-tissue contrast resolution compared to plain radiographs. Contrast enhancement improves vascular and parenchymal visualisation.

Study Type	Key Indications	Contrast
CT Head (non-contrast)	Haemorrhage, trauma, acute stroke exclusion	None
CT Pulmonary Angiogram (CTPA)	Pulmonary embolism, pulmonary vascular disease	IV contrast
CT Abdomen/Pelvis	Appendicitis, bowel obstruction, masses, AAA	Oral ± IV
CT Coronary Angiogram	CAD assessment, anomalous coronaries	IV contrast
CT Spine	Trauma, disc herniation, cord compression	Usually none

4. Clinical Procedures Overview

4.1 Venepuncture & IV Access

Peripheral venous cannulation is one of the most frequently performed clinical procedures. Correct technique minimises complications including haematoma, phlebitis, and extravasation.

- Site selection: antecubital fossa (large veins), dorsum of hand, forearm
- Aseptic technique: Hand hygiene, tourniquet, skin antisepsis with 70% alcohol, allow to dry 30 s
- Insertion angle: 15–30° bevel up; advance after flashback; remove tourniquet before flushing
- Flush with 5–10 mL 0.9% saline; secure and date cannula
- Replace peripheral cannulas within 72–96 hours or when clinically indicated

4.2 Urinalysis Interpretation

Finding	Possible Cause
Leucocyte esterase +ve	Urinary tract infection, urethritis
Nitrites +ve	Gram-negative bacteriuria (E. coli, Klebsiella)
Blood (haematuria)	UTI, renal calculus, trauma, malignancy, GN
Protein (proteinuria)	Glomerulonephritis, nephrotic syndrome, hypertensive nephropathy
Glucose (glycosuria)	Diabetes mellitus (when BGL > renal threshold ~10 mmol/L)
Ketones	DKA, starvation, alcoholic ketoacidosis
Bilirubin +ve	Hepatocellular disease, obstructive jaundice

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